

Notes: Adrian, Etula, and Muir (JF, 2014)

Financial Intermediaries and the Cross-Section of Asset Returns

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1 Big picture

- **Question.** Can the marginal value of wealth of financial intermediaries price the cross-section of asset returns better than the marginal value of the representative household?
- **Core idea.** If broker-dealers are the marginal investors in many markets, their balance-sheet condition should be embedded in the stochastic discount factor (SDF).
- **Empirical SDF proxy.** The paper uses quarterly shocks to U.S. securities broker-dealer leverage as a proxy for intermediary marginal value of wealth.
- **Economic intuition.** When funding conditions tighten, broker-dealers delever. Their marginal value of wealth is high. Assets that perform poorly when broker-dealer leverage falls are risky and must offer high expected returns.
- **Main cross-section.** The main test assets are 25 size/book-to-market portfolios, 10 momentum portfolios, and six Treasury bond portfolios sorted by maturity.
- **Main result.** A one-factor intermediary leverage model explains roughly 77% of the cross-sectional variation in average excess returns, with average absolute pricing error around 1.3% per year.
- **Why this is surprising.** The single leverage factor performs similarly to multi-factor benchmarks designed to price these assets, including Fama-French, momentum, and a bond factor.
- **Leverage factor interpretation.** Broker-dealer leverage shocks are procyclical: they are positively related to broker-dealer asset growth and financial-sector returns and negatively related to market volatility and credit spreads.
- **Robustness.** The factor prices separate equity portfolios, momentum portfolios, Treasury bond portfolios, beta-sorted portfolios, and leverage-beta-sorted portfolios. Random-factor simulations rarely replicate its pricing performance.
- **Tradable version.** The authors build a leverage-mimicking portfolio (LMP) by projecting the nontraded leverage factor onto traded benchmark returns. The LMP performs well in monthly tests back to 1936 and has a high Sharpe ratio.
- **Economic takeaway.** Intermediary balance sheets can summarize time-varying market-wide funding conditions. Cross-sectional expected returns compensate investors for exposure to intermediary funding shocks.

2 Data and factor construction

2.1 Why broker-dealers?

- Broker-dealers trade frequently, participate in many asset classes, have sophisticated risk-management systems, and adjust balance sheets actively.
- These features make them plausible marginal investors in many securities markets.
- The paper therefore shifts focus from the representative household SDF to an intermediary SDF.

Interpretation: The empirical strategy is not to claim that every investor is a broker-dealer. The claim is that the pricing kernel relevant for many traded assets can be proxied by the balance-sheet condition of active intermediaries.

2.2 Broker-dealer balance-sheet data

- The main data source is the Federal Reserve Flow of Funds, Table L.129.
- Flow of Funds data are supplemented and interpreted using SEC FOCUS reports.

- The sample begins in 1968Q1 because earlier broker-dealer equity data are unreliable.
- The main sample ends in 2009Q4.
- Assets include credit-market instruments, equities, security credit, and a large category of miscellaneous assets.
- Liabilities include net repo, corporate and foreign bonds, trade payables, security credit, taxes payable, and miscellaneous liabilities.

Interpretation: Table I is important because it shows that broker-dealer balance sheets are not simply household balance sheets. Broker-dealers hold and finance positions through wholesale market instruments.

2.3 Leverage factor

Broker-dealer leverage is assets divided by equity. The leverage factor is the seasonally adjusted log change in broker-dealer leverage:

$$\text{LevFac}_t = [\Delta \ln (\text{Leverage}_t^{BD})]^{SA}. \quad (1)$$

- Seasonal adjustment is done with quarterly seasonal dummies in real time.
- The factor is a shock to leverage, not the level of leverage.
- Negative leverage shocks correspond to deleveraging and tight funding conditions.

Interpretation: Because broker-dealer leverage is persistent, log changes are close to innovations. The factor is designed to measure changes in intermediary balance-sheet capacity.

2.4 Funding-condition proxies

- The leverage factor is positively correlated with broker-dealer asset growth.
- It is negatively correlated with market volatility and the Baa-Aaa credit spread.
- It is positively correlated with financial-sector stock returns.
- These signs are consistent with the view that declines in leverage reflect bad funding states.

Interpretation: The correlation evidence is a mechanism check: it connects the statistical factor to funding conditions rather than treating it as a black-box return predictor.

2.5 Test assets

The main cross-section contains 41 portfolios:

- 25 size and book-to-market portfolios;
- 10 momentum portfolios;
- six Treasury bond portfolios sorted by maturity.

Additional tests use:

- equity-only and bond-only subsets;
- a tradable leverage mimicking portfolio;
- the Fama-French benchmark factors;
- portfolios sorted on leverage betas;
- “betting against beta” portfolios.

Interpretation: Using many test assets across equities and bonds addresses the concern that one factor merely fits a narrow set of portfolios.

3 Models and empirical specifications

3.1 Intermediary SDF

The model proposes an SDF affine in the broker-dealer leverage factor:

$$m_t = 1 - b\text{LevFac}_t. \quad (2)$$

For excess returns, the pricing condition is

$$E[m_t R_{i,t}^e] = 0. \quad (3)$$

Interpretation: If the coefficient b is positive, assets that covary negatively with leverage shocks pay off badly when intermediary marginal value is high and must have high expected returns.

3.2 Single-factor cross-sectional model

The implied expected-return relation is

$$E[R_i^e] = \beta_{i,Lev} \lambda_{Lev}, \quad (4)$$

where

$$\beta_{i,Lev} = \frac{\text{Cov}(R_{i,t}^e, \text{LevFac}_t)}{\text{Var}(\text{LevFac}_t)}. \quad (5)$$

Interpretation: The paper's main one-factor model says that assets with higher exposure to broker-dealer leverage shocks should have higher average excess returns.

3.3 First-pass time-series regressions

For a general factor vector f_t , the first-pass regression is

$$R_{i,t}^e = c_i + \beta_i' f_t + \varepsilon_{i,t}, \quad i = 1, \dots, N. \quad (6)$$

Interpretation: These regressions estimate each asset's exposure to the factor or factor set.

3.4 Second-pass cross-sectional regressions

The second pass estimates factor prices of risk:

$$E[R_i^e] = a + \beta_i' \lambda + \xi_i. \quad (7)$$

The paper reports:

- prices of risk and Fama-MacBeth/Shanken t -statistics;
- mean absolute pricing error, MAPE;
- adjusted R^2 and confidence intervals;
- a χ^2 test of whether pricing errors are jointly zero.

Interpretation: A successful model has a small intercept, statistically meaningful factor price, low pricing errors, and high adjusted R^2 across the test assets.

3.5 Benchmarks

The leverage model is compared with:

- CAPM;
- Fama-French three-factor model;
- Fama-French plus momentum;
- Fama-French plus momentum plus the first principal component of the yield curve;
- leverage plus market return.

Interpretation: These benchmarks are chosen because they are tailored to the portfolios being priced: equity factors for stock portfolios, momentum for momentum portfolios, and a bond factor for Treasury portfolios.

3.6 Leverage mimicking portfolio

Because the broker-dealer leverage factor is nontraded and quarterly, the paper constructs a tradable leverage mimicking portfolio (LMP) by projecting LevFac_t onto benchmark traded factors:

$$\text{LevFac}_t = a + \boldsymbol{\theta}' \mathbf{R}_t + \omega_t, \quad (8)$$

where $\mathbf{R}_t$ contains traded factor returns. The fitted component is

$$\text{LMP}_t = \boldsymbol{\theta}' \mathbf{R}_t. \quad (9)$$

Interpretation: The LMP allows monthly tests back to 1936 and mean-variance analysis, but it is mechanically a portfolio formed from benchmark traded factors.

3.7 Leverage beta sorts and betting-against-beta portfolios

The paper also studies portfolio sorts using estimated exposures to the leverage factor:

$$R_{i,t}^e = c_i + \beta_{i,Lev} \text{LevFac}_t + \varepsilon_{i,t}. \quad (10)$$

The beta-sorted portfolios test whether high-leverage-beta stocks earn higher average returns. The betting-against-beta portfolios test whether the leverage factor is related to funding constraints.

Interpretation: These tests are useful because they move beyond the original 41 portfolios and examine broader equity return patterns connected to intermediary funding.

4 Identification and empirical design

4.1 What the paper identifies

- This is an empirical asset-pricing paper, not a causal experiment.
- The main object is whether exposures to intermediary leverage shocks price the cross-section of average returns.
- The identifying variation is cross-sectional: assets differ in their covariance with the leverage factor.
- The factor is time-series variation in broker-dealer leverage growth.

Interpretation: The relevant test is not whether broker-dealer leverage causes individual asset returns, but whether leverage exposure is a priced risk.

4.2 Why the factor is economically interpretable

- Broker-dealer leverage is procyclical, unlike household leverage.
- Leverage shocks are positively correlated with asset growth and financial-sector stock returns.
- Leverage shocks are negatively correlated with market volatility and credit spreads.
- These facts line up with funding-constraint theories of intermediary asset pricing.

Interpretation: A key advantage of the factor is that it is connected to intermediary balance-sheet capacity, not just selected to maximize fit.

4.3 Addressing spurious-factor concerns

- Lewellen, Nagel, and Shanken caution that high R^2 can arise mechanically in asset-pricing tests with many portfolios.
- The authors use many test assets and a one-factor model, which makes the performance harder to dismiss.
- Simulation tests draw random factors and ask how often they match the leverage factor's R^2 and intercept performance.
- The probabilities are extremely small.

Interpretation: The simulation exercise asks whether the leverage factor is just a lucky high- R^2 factor. Table VIII says it is unlikely.

4.4 Role of the tradable LMP

- The LMP is useful because it is a traded return.
- It permits monthly tests and a sample going back to 1936.
- It also permits mean-variance analysis against benchmark factors.
- The limitation is that the LMP is constructed from returns, so it cannot fully replace the original balance-sheet factor.

Interpretation: The nontraded leverage factor is the economic object. The LMP is a robustness tool and a way to connect the factor to standard portfolio analysis.

4.5 What the paper cannot fully prove

- The factor is measured from aggregate broker-dealer balance sheets, not from every marginal investor's shadow value.
- Quarterly Flow of Funds data may contain measurement error.
- The analysis does not estimate a structural intermediary asset-pricing model.
- The leverage factor is not a traded factor in its original form.
- The empirical evidence supports the intermediary-SDF interpretation but does not uniquely identify one theory.

Interpretation: The results are strongest as evidence that intermediary balance sheets contain priced information. They do not fully settle the microfoundation of that price of risk.

5 Tables and figures

This section keeps the compact visual style used in the previous notes. Adjacent original tables and figures are grouped together to save space and improve reading speed.

5.1 Figure 1, Table I, and Table II: the factor and balance-sheet data

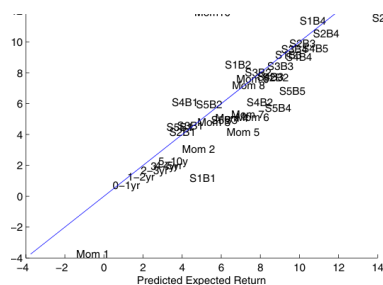
Read: Figure 1.

- The predicted and realized average returns lie close to the 45-degree line.
- The leverage factor prices 25 size/book-to-market portfolios, 10 momentum portfolios, and six Treasury bond portfolios in one cross-section.
- The figure visually summarizes the paper's main pricing claim.

Read: Table I and Table II.

- Table I decomposes broker-dealer assets and liabilities from Flow of Funds and FOCUS report categories.
- Table II shows the leverage factor is strongly correlated with broker-dealer asset growth, 0.73.
- It is negatively correlated with market volatility, -0.37 , and the Baa-Aaa spread, -0.16 .
- It is positively correlated with financial-sector stock returns, 0.18.

Economic message: The factor is not just a statistical construction. It reflects broker-dealer balance-sheet expansion and contraction, and it comoves with standard indicators of funding conditions.



realized versus predicted mean returns: leverage factor. We plot the returns of 35 equity portfolios (25 size- and book-to-market-sorted portfolios) and six Treasury bond portfolios (sorted by maturity) against the leverage factor. The leverage factor is constructed as the cross-section return predicted by our single-factor financial intermediary leverage factor (with an intercept $E[R^*] = \beta_{int} \lambda_{int}$). The sample period is 1968Q1 to 2009Q4, but returns are expressed in percent per year.

Figure 1: leverage factor predicted vs. realized returns

Table I: broker-dealer balance sheet

Assets from Flow of Funds (billions)		Liabilities from Flow of Funds (billions)	
cash (including segregated cash)	\$96.9	Net repo	\$404.7
debt market instruments	\$557.6	Corporate and foreign bonds	\$129.7
commercial paper	\$36.2	Trade payables	\$18.1
treasury securities (net of shorts)	\$94.5	Security credit	\$936.6
government securities	\$149.8	Taxes payable	\$3.6
uncollateralized securities	\$40.0	Miscellaneous liabilities*	\$480.7
corporate and foreign bonds	\$185.6	Payables to brokers and dealers	
other (syndicated loans, etc.)	\$51.4	Securities sold not yet purchased	
corporate equities	\$117.2	Payables	
security credit	\$278.2	Subordinated liabilities	
miscellaneous assets*	\$1,025.3		
Receivables			
Reverse repos			
Options and arbitrage			
Commodities			
Investments not readily marketable			
Securities borrowed			
Secured demand notes			

Table I: broker-dealer balance sheet

Broker-Dealer Leverage Is Procyclical
This table presents the correlation of U.S. broker-dealer leverage growth with a selection of variables, including the log asset growth of U.S. broker-dealers, market volatility (constructed quarterly using weekly data of the value-weighted market return), the Baa-Aaa spread, and the blue-weighted stock return of the U.S. financial sector. The sample is 1968Q1 to 2009Q4.

Correlation of Broker-Dealer Leverage Factor with:			
Log Broker-Dealer Asset Growth	Market Volatility	Baa-Aaa Spread	Financials Stock Return
0.73	-0.37	-0.16	0.18
t-value	0.00	0.00	0.02

Table II: leverage factor correlations

5.2 Figure 2 and Figure 3: intermediary leverage dynamics

Read:

- Figure 2 plots standardized broker-dealer log leverage and leverage shocks.
- Large negative shocks appear around major financial stress episodes, including the 1987 crash and the 2008 crisis.
- Figure 3 contrasts households with broker-dealers.
- Household leverage growth is negatively related to asset growth, consistent with passive balance-sheet mechanics.

- Broker-dealer leverage growth is positively related to asset growth, showing active procyclical balance-sheet management.

Economic message: Procyclical intermediary leverage is the mechanism behind the factor: when balance-sheet capacity expands, leverage rises; when constraints tighten, broker-dealers delever.

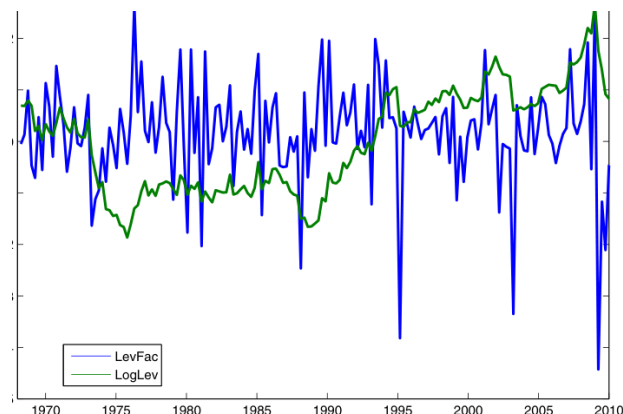


Figure 2. Log-leverage and leverage factor. We plot the log-leverage and the leverage factor (changes in log-leverage) of security broker-dealers, 1968Q1 to 2009Q4. We standardize each series
Figure 2: broker-dealer log leverage and leverage factor

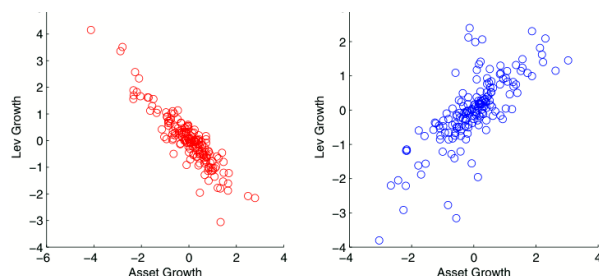


Figure 3. Cyclical relationship of leverage. We plot leverage growth versus asset growth for households
Figure 3: household versus broker-dealer leverage cyclical relationship

5.3 Table III, Figure 4, and Figure 5: main asset-pricing results and benchmarks

Read: Table III.

- The leverage factor price of risk is positive and statistically significant: 62.21 with Fama-MacBeth $t = 4.62$ and Shanken $t = 3.12$.
- The cross-sectional intercept is only 0.12% per year.
- The total MAPE is 1.31% per year.
- The adjusted R^2 is 77% across 41 assets.
- The leverage factor performs close to the five-factor benchmark despite using only one factor.

Read: Figures 4 and 5.

- Figure 4 shows the predicted-realized relation for Fama-French factors.
- Figure 5 shows the five-factor benchmark.
- Compared with these benchmarks, the single leverage factor is visually competitive.

Economic message: The main empirical message is that a one-factor intermediary leverage model prices a broad equity-bond cross-section nearly as well as tailored multifactor models.

Table III
Pricing the Size, Book-to-Market, Momentum, and Bond Portfolios

This table presents pricing results for the 25 size and book-to-market, 10 momentum, and 6 Treasury bond portfolios sorted by maturity. Each model is estimated as $E[R^e] = \lambda_0 + \beta_{\text{size}}\lambda_{\text{size}} + \beta_{\text{b/m}}\lambda_{\text{b/m}} + \beta_{\text{mom}}\lambda_{\text{mom}} + \beta_{\text{PC1}}\lambda_{\text{PC1}} + \beta_{\text{Lev}}\lambda_{\text{Lev}}$. FF denotes the Fama-French three factors, Mom the momentum factor, PC1 the first principal component of the yield curve, and LevFac our leverage factor. Panel A reports the prices of risk with Fama-MacBeth and Shanken t -statistics. Panel B reports test diagnostics, including mean absolute pricing errors (MAPEs) by portfolio group, adjusted R^2 s with corresponding confidence intervals (C.I.), and a χ^2 statistic that tests whether the pricing errors are jointly zero. $E[R^e]$ gives the average excess return to be explained. Data are quarterly, 1968Q1 to 2009Q4. Returns and risk premia are reported in percent per year (quarterly percentages multiplied by four).

Panel A: Prices of Risk						
	CAPM	FF	FF, Mom	FF, Mom, PC1	LevFac	LevMkt
Intercept	3.39	3.16	1.06	0.66	0.12	-0.19
t -FM	3.55	4.09	1.51	1.14	0.06	-0.21
t -Shanken	3.54	4.03	1.34	1.01	0.04	-0.14
LevFac					62.21	60.97
t -FM					4.62	5.29
t -Shanken					3.12	3.65
Mkt	3.06	2.30	4.54	4.89		5.46
t -FM	0.99	0.80	1.59	1.71		1.75
t -Shanken	0.99	0.80	1.58	1.70		1.55
SMB		1.76	1.57	1.63		
t -FM		0.93	0.83	0.87		
t -Shanken		0.93	0.82	0.86		
HML		3.33	4.37	4.34		
t -FM		1.45	1.90	1.89		
t -Shanken		1.45	1.86	1.85		
MOM			7.82	7.75		
t -FM			2.94	2.91		
t -Shanken			2.92	2.89		
PC1				14.99		
t -FM				1.03		
t -Shanken				0.93		

Panel B: Test Diagnostics							
MAPE	$E[R^e]$	CAPM	FF	FF, Mom	FF, Mom, PC1	LevFac	LevMkt
Size B/M	7.86	2.62	1.81	1.05	1.01	1.16	1.11
MOM	5.00	0.05	0.75	1.47	1.40	1.00	1.05

Table III: pricing size/book-to-market, momentum, and bond portfolios

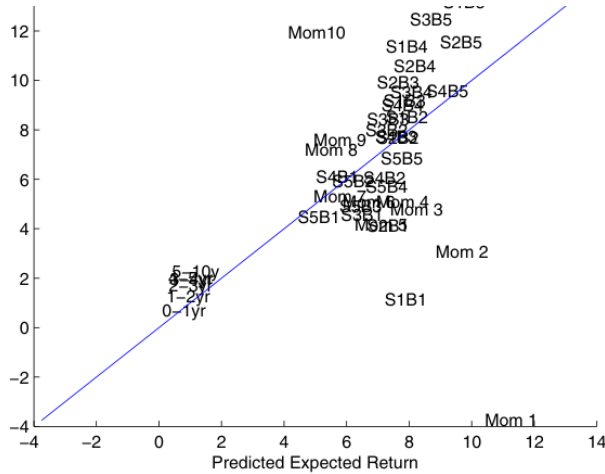


Figure 4: Fama-French predicted vs. realized returns

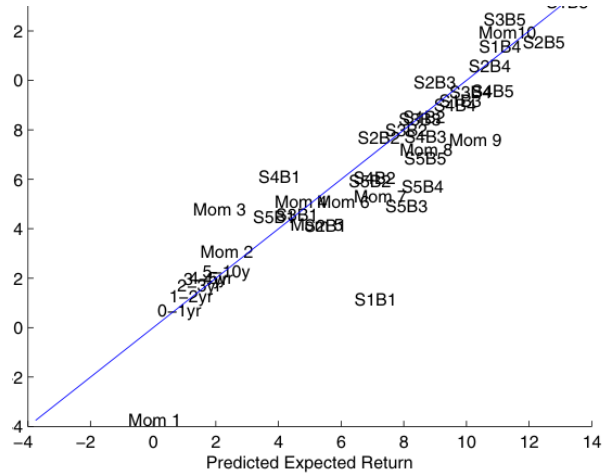


Figure 5: five-factor benchmark predicted vs. realized returns

5.4 Table IV, Table V, and Table VI: pricing errors by portfolio group

Read:

- Table IV reports individual pricing errors for benchmark models and the leverage factor model.
- The leverage model's main outlier is the highest momentum portfolio, but many portfolios have small errors.
- Table V focuses on equity portfolios and shows that the leverage factor performs well on size/book-to-market

and momentum portfolios separately.

- Table VI focuses on Treasury bond portfolios and shows that the leverage factor explains a large fraction of the bond return cross-section.

Economic message: The pricing performance is not driven by one asset class. The leverage factor works across equities and bonds, which supports an intermediary-wide SDF interpretation.

Table IV
Pricing Errors: Pricing the Size, Book-to-Market, Momentum, and Bond Portfolios

This table presents pricing errors ($E[R^*] - \lambda_0 - \beta_{\text{Mkt}}\lambda_{\text{Mkt}} - \beta_{\text{SMB}}\lambda_{\text{SMB}} - \beta_{\text{HML}}\lambda_{\text{HML}} - \beta_{\text{Mom}}\lambda_{\text{Mom}} - \beta_{\text{Bond}}\lambda_{\text{Bond}}$) in percent per year for our leverage factor model and five-factor benchmark (Mkt, SMB, HML, Mom, and Bond) corresponding directly to the previous table. Panel B reports cross-sectional results when we drop the two most mispriced portfolios: the small growth portfolio (S1B1) and the past winners portfolio (Mom10).

Panel A: Individual Pricing Errors						
Size and Book-to-Market Portfolios						
	Low	Book-to-Market			High	
$E[R^*]$: Average Returns						
Small	1.15	8.52	9.17	11.38	13.16	
	4.14	7.67	9.91	10.55	11.54	
Size	4.56	8.00	8.43	9.54	12.46	
	6.09	6.07	7.72	9.01	9.57	
Big	4.47	5.93	4.93	5.72	6.83	
Leverage Pricing Errors						
Small	-3.27	2.31	0.43	1.40	1.39	
	0.73	-0.42	0.46	-0.10	-2.10	
Size	0.76	0.79	0.09	0.49	1.88	
	2.57	-1.24	-0.11	-0.27	-0.53	
Big	1.20	1.18	-0.59	-2.51	-2.11	
Five-Factor Pricing Errors						
Small	-5.26	0.47	-0.05	0.89	0.78	
	-0.64	0.97	1.46	0.50	0.00	
Size	0.52	0.48	0.43	0.02	1.88	
	2.76	-0.43	-0.28	0.03	-0.45	
Big	1.30	-0.18	-2.32	-2.14	-1.06	
Momentum Portfolios				Bond Portfolios		
	$E[R^*]$	LevFac	5-Fac		$E[R^*]$	LevFac
Mom 1	-3.75	-2.46	-2.96	0-1 yr	0.70	0.15
Mom 2	3.06	-1.00	1.71	1-2 yr	1.28	-0.56
Mom 3	4.80	-0.03	3.52	2-3 yr	1.70	-0.02
Mom 4	5.10	-0.63	1.02	3-4 yr	1.95	-0.27
Mom 5	4.15	-2.15	-0.27	4-5 yr	2.00	-0.52
Mom 6	5.11	-1.67	-0.10	5-10 yr	2.29	-0.70
Mom 7	5.29	-1.23	-1.23			
Mom 8	7.19	0.63	-0.57			
Mom 9	7.61	0.81	-1.61			
Mom 10	11.93	7.25	1.81			

(Continued)

Table IV—Continued

Panel B: Pricing Results Dropping Largest Pricing Error Portfolios								
	All Portfolios		Drop Small Growth		Drop Mom 10		Drop Both	
	5-Fac	LevFac	5-Fac	LevFac	5-Fac	LevFac	5-Fac	LevFac
MAPE	1.00	1.20	0.78	1.13	0.94	1.01	0.74	0.93
AdjR ²	0.81	0.77	0.88	0.79	0.82	0.87	0.89	0.87
T ² (y ²)	110.19	67.87	72.42	42.95	85.15	45.78	55.27	27.40
p-value	0.0%	0.0%	0.0%	26.8%	0.0%	18.1%	0.9%	87.5%

We find this ratio to be about six, making the price of traded leverage risk about 10% per annum—a number much more in line with standard traded factors. Again, it is crucial to understand that the presence of noise like ω_t will not impact the cross-sectional results in any way since those rely solely

Table IV: pricing errors by portfolio

Table V
Equity Portfolios

This table presents pricing results for the 25 size and book-to-market (left panel) and 25 size and momentum portfolios (right panel). Each model is estimated as $E[R^e] = \lambda_0 + \beta_{fac}\lambda_{fac}$. FF denotes the Fama-French three factors, Mom the momentum factor, and LevFac our leverage factor. Panel A reports the estimated risk premia, along with Fama-MacBeth and Shanken t -statistics, respectively. Panel B reports test diagnostics, including MAPEs, the largest absolute pricing error (MAX), adjusted R^2 s with corresponding confidence intervals (C.I.), and a χ^2 statistic testing whether the pricing errors are jointly zero. $E[R^e]$ gives the average excess return to be explained. Data are quarterly, 1968Q1 to 2009Q4. Returns and risk premia are reported in percent per year (quarterly percentages multiplied by four).

Panel A: Prices of Risk								
25 Size and Book-to-Market Portfolios				25 Size and Momentum Portfolios				
	CAPM	FF	LevFac	LevMkt	CAPM	FF, Mom	LevFac	LevMkt
Intercept	12.11	15.58	1.00	0.27	3.51	11.72	0.31	-5.47
t -FM	2.99	3.84	0.25	0.06	3.41	1.72	0.07	-0.96
t -Shanken	2.97	3.57	0.18	0.04	3.37	1.60	0.04	-0.53
LevFac			55.78	56.80			69.66	81.82
t -FM			3.30	3.27			3.66	4.25
t -Shanken			2.34	2.29			2.28	2.38
Mkt	-3.81	-10.19		5.20	-5.88	-4.76		10.62
t -FM	-0.80	-2.09		0.95	-1.17	-0.64		1.69
t -Shanken	-0.79	-1.98		0.70	-1.16	-0.60		1.00
SMB		1.85				2.39		
t -FM		0.98				1.12		
t -Shanken		0.97				1.10		
HML		5.76				-4.01		
t -FM		2.42				-1.00		
t -Shanken		2.38				-0.95		
MOM						8.40		
t -FM						3.19		
t -Shanken						3.18		

Panel B: Test Diagnostics								
25 Size and Book-to-Market Portfolios				25 Size and Momentum Portfolios				
MAPE:	CAPM	FF	LevFac	LevMkt	CAPM	FF, Mom	LevFac	LevMkt
Intercept	12.11	15.58	1.00	0.27	3.51	11.72	0.31	-5.47
Total	14.41	16.69	2.09	1.34	6.48	12.83	2.47	7.57
MAX	5.71	4.33	3.72	3.95	9.99	4.54	7.01	6.06

Table V: equity portfolios

Table VI
Treasury Bonds

This table presents pricing results for the six Treasury bond portfolios sorted by maturity. Each model is estimated as $E[R^e] = \beta_{fac}\lambda_{fac}$, without an intercept. FF denotes the Fama-French three factors, Mom the momentum factor, PC1 the first principal component of the yield curve, and LevFac our leverage factor. For the traded factors, we report time series α s as pricing errors, since there are only six test assets. The last column, LevFac NRE (not reestimated), gives pricing errors for the leverage factor using the price of risk from the larger cross-section of stocks and bonds. Panel B reports the estimated risk premia, along with Fama-MacBeth and Shanken t -statistics, respectively. Panel C reports test diagnostics, including MAPEs, the largest pricing error (MAX), adjusted R^2 s with corresponding confidence intervals (C.I.), and a χ^2 statistic testing whether the pricing errors are jointly zero. $E[R^e]$ gives the average excess return to be explained. Data are quarterly, 1968Q1 to 2009Q4. Returns and prices of risk are reported in percent per year (quarterly percentages multiplied by four).

Panel A: Pricing Errors							
	$E[R^e]$	CAPM	FF	FF, Mom	PC1	LevFac	Lev NRE
0-1 yr	0.70	0.65	0.61	0.58	0.36	0.33	0.27
1-2 yr	1.28	1.16	1.08	0.84	0.31	0.28	0.10
2-3 yr	1.70	1.54	1.47	1.04	0.22	0.12	-0.15
3-4 yr	1.95	1.77	1.73	1.13	0.11	-0.05	-0.40
4-5 yr	2.00	1.84	1.85	1.03	-0.13	-0.19	-0.58
5-10 yr	2.29	2.01	2.11	0.96	-0.27	-0.03	-0.44

Panel B: Prices of Risk				
	PC1	LevFac	Lev NRE	LevMkt
LevFac		52.90	62.21	40.14
t -FM		2.28	NA	2.05
t -Shanken		1.65	NA	1.57
PC1	31.52			18.88
t -FM	2.27			1.95
t -Shanken	2.14			1.50

Panel C: Test Diagnostics				
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Table VI: Treasury bond portfolios

5.5 Table VII and Table VIII: time-series regressions and simulation robustness

Read: Table VII.

- The table reports time-series regressions of test-asset excess returns on the leverage factor.
- Bond portfolios have especially strong leverage betas and high time-series explanatory power.
- Equity portfolios also show economically meaningful exposure patterns.

Read: Table VIII.

- Random-factor simulations ask whether a noise factor could replicate the leverage factor's performance.
- The probability of matching both high R^2 and low intercept is essentially zero in the main sample.
- Robustness across alternative starting dates also supports the interpretation.

Economic message: These tables address two critiques: whether the factor has meaningful time-series exposure and whether the cross-sectional fit could be spurious.

Table VII
Time-Series Regressions

This table presents time-series regressions of excess returns on the leverage factor $R_{L,F}^e = c_L + \beta_{L,F} LevF \alpha_{c_L} + \epsilon_{L,F}$. Data are quarterly, 1968Q1 to 2009Q4. Returns are reported in percent per year (quarterly percentages multiplied by four). Leverage betas are multiplied by 100 for convenience.

Size and Book-to-Market Portfolios									
	Low	Book-to-Market		High					
$E[R^e]$: Average (Annualized) Returns									
Small	1.15	8.52	9.17	11.38	13.16				
	4.14	7.67	9.91	10.55	11.54				
Size	4.56	8.00	8.43	9.54	12.46				
	6.09	6.07	7.72	9.01	9.57				
Big	4.47	5.93	4.93	5.72	6.83				
$\beta_{L,F}$: Leverage Betas ($\times 10^{-2}$)									
Small	6.92	9.79	13.86	15.85	18.72				
	5.29	12.82	15.01	16.93	21.73				
Size	5.92	11.41	13.22	14.35	16.81				
	5.48	11.55	12.41	14.72	16.04				
Big	5.06	7.45	8.68	13.03	14.19				
t -stats									
Small	0.72	1.21	1.93	2.33	2.44				
	0.62	1.79	2.40	2.78	3.18				
Size	0.76	1.81	2.32	2.53	2.67				
	0.78	1.93	2.23	2.70	2.55				
Big	0.93	1.51	1.90	2.81	2.79				
R^2 : R-square									
Small	0.31%	0.87%	2.18%	3.14%	3.43%				
	0.23%	1.89%	3.32%	4.43%	5.73%				
Size	0.34%	1.91%	3.12%	3.70%	4.10%				
	0.36%	2.17%	2.89%	4.17%	3.74%				
Big	0.52%	1.34%	2.12%	4.52%	4.46%				
Momentum Portfolios				Bond Portfolios					
	$E[R^e]$	$\beta_{L,F}$	t -stat	R^2	$E[R^e]$	$\beta_{L,F}$	t -stat	R^2	
Mom 1	-3.75	-2.26	-0.24	0.03%	0-1yr	0.70	0.69	1.89	2.63%
Mom 2	3.06	6.32	0.85	0.43%	1-2yr	1.28	1.89	1.46	2.56%
Mom 3	4.80	7.56	1.23	0.89%	2-3yr	1.70	2.98	2.04	3.30%
Mom 4	5.10	9.01	1.64	1.58%	3-4yr	1.95	3.79	2.06	3.58%
Mom 5	4.15	9.94	1.98	2.30%	4-5yr	2.00	4.14	2.00	3.82%

Table VII: time-series regressions

Table VIII
Robustness

Panel A shows robustness of our main measure to several starting dates. End dates are 2009Q4 except the last column, which excludes the financial crisis, ending in 2005Q4. Panel B examines how likely a "noise" factor could produce our results. We run 100,000 simulations where we draw randomly from the empirical distribution of our leverage factor. We rerun all our tests, including constructing the LMP (leverage mimicking portfolio). For each statistic, we report the probability that the random factor does as well as our leverage factor (i.e., the probability the random factor has an R^2 as large as our leverage factor or an intercept as small as our leverage factor, or both jointly in the "Joint" column).

Panel A: Robustness to Various Start Dates								
Years	1966	1967	1968	1969	1970	1971	1972	1968 to 2005
Intercept	0.46	0.37	0.12	0.19	0.67	1.53	1.29	-0.37
Adj R^2	0.72	0.74	0.77	0.77	0.79	0.76	0.79	0.67
$T^2(\chi^2)$	60.49	64.20	67.87	72.00	74.72	69.07	61.83	67.61
p -value	2%	1%	0%	0%	0%	0%	1%	0%
Panel B: What Are the Odds a Random Factor Could Produce Our Results?								
	R^2		MAPE		Intercept		Joint R^2 -Int	
Noise Factor	p -value		0.01%		0.00%		0.19%	
	R^2		MAPE		Sharpe		Joint R^2 -Sharpe	
Noise LMP	p -value		0.18%		0.00%		1.63%	

Table VIII: simulation robustness

5.6 Table IX, Table X, Figure 6, and Table XI: the tradable LMP

Read: Tables IX and X.

- Table IX constructs and tests the leverage mimicking portfolio (LMP).
- The LMP performs comparably to Fama-French and momentum benchmarks in time-series and cross-sectional tests.
- Table X shows that the LMP can price benchmark factors well, indicating that it inherits important pricing information.

Read: Figure 6 and Table XI.

- Figure 6 places the LMP near the sample mean-standard deviation frontier.
- Table XI reports a monthly LMP Sharpe ratio of 0.29 and an annualized Sharpe ratio of 0.99.
- These Sharpe ratios are higher than those of the market, SMB, HML, and momentum factors individually.

Economic message: The LMP evidence shows that the leverage factor is not only a quarterly nontraded statistic. Its projection onto traded returns has strong mean-variance properties and pricing content.

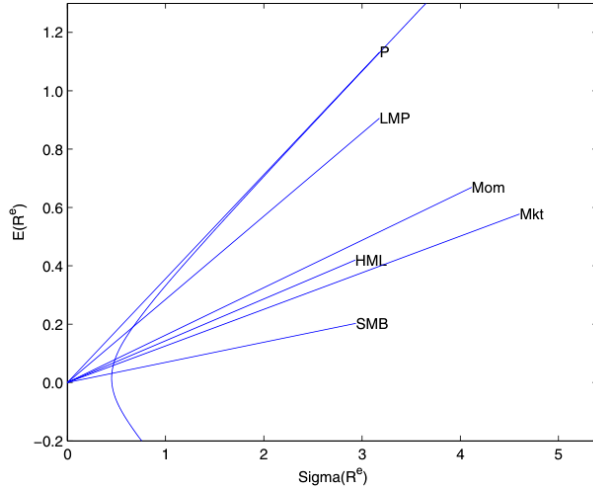
Table IX
The LMP: Comparing Models

This table presents the time-series alphas generated by each model (LMP, Fama-French, and Fama-French plus momentum). MAPE represents the mean absolute pricing error given in percent per annum ($MAPE = \frac{1}{N} \sum_{i=1}^N |\alpha_i|$). SBM represents the 25 size and book-to-market portfolios, MOM the 10 momentum portfolios, and Bond the six bond portfolios. The first column (MEAN) gives the absolute average return to be explained. We also report the GRS F -statistic that the alphas are jointly zero and its associated p -value. The sample period is 1968Q1 to 2009Q4. The last panel compares cross-sectional results using each of the factor models. We confirm our pricing results using monthly data going back to 1936. For the 1936 to 2009 sample, we only use the size, book-to-market, and momentum portfolios, since bond data are not available.

Panel A: Time-Series Alphas				
MAPE	Mean	LMP	FF, MOM	FF
SBM	7.86	1.15	1.04	1.57
MOM	5.80	1.66	1.46	4.36
Bond	3.04	0.59	0.93	1.47
Total	6.33	1.19	1.13	2.24
Model Fit				
		LMP	FF, MOM	FF
GRS		2.57	2.28	4.48
p -value		0	0	0
Panel B: Cross-Sectional Results across Time Periods				
Time Period		LMP	FF, MOM	FF
1968 to 2009,	Intercept	-0.32	1.06	3.12
Quarterly	Adj R^2	0.78	0.81	0.16
1936 to 2009,	Intercept	-3.00	14.74	27.97
Monthly	Adj R^2	0.63	0.81	0.52

benchmark on an R^2 basis. Our empirical results thus continue to hold over the longer time span and at the higher monthly frequency, providing out-of-sample robustness tests of our results. Overall, the LMP alone performs comparably to the Fama-French and momentum factors.

Table IX: LMP comparison models



6. Risk-return for LMP and benchmark pricing factors. The hyperbola plot shows the mean-standard deviation frontier ("the efficient frontier") for the six Fama-French factors, the Fama-French factors (Mkt, SMB, and HML), and the momentum factor.

Figure 6: risk-return frontier and LMP

5.7 Table XII and Table XIII: leverage beta and betting-against-beta portfolios

Read: Table XII.

- Stocks sorted on prereanking leverage betas earn higher average returns as leverage beta rises.
- In the three-portfolio sort, the high-minus-low return is about 2.82% per year.
- In the five-portfolio sort, the high-minus-low return is about 3.87% per year.

Read: Table XIII.

- Betting-against-beta portfolios are connected to funding constraints.
- Levered low-beta portfolios have higher average returns and stronger leverage betas.

Table X
Pricing the Benchmark Factors with the LMP

This table presents time-series regressions of the four benchmark factors, market, SMB, HML, and momentum, on the LMP. We report the quarterly results for our main sample 1968 to 2009, as well as monthly results that begin in 1936. We provide the average mean return to be explained, along with the time-series regression statistics. Mean returns and alphas are reported in annual percentage terms for consistency.

Model: $R_t^f = \alpha + \beta LMP_t + \varepsilon_t$						
Quarterly Data: 1968 to 2009						
	$E[R^f]$	Alpha	T-Alpha	Beta	T-Beta	R^2
Mkt	5.44	0.08	(0.03)	0.52	(2.90)	0.10
SMB	3.07	1.72	(0.75)	0.13	(1.19)	0.02
HML	4.06	-0.22	(-0.74)	0.44	(2.92)	0.12
Mom	7.99	4.24	(1.08)	0.36	(1.85)	0.06
Monthly Data: 1936 to 2009						
	$E[R^f]$	Alpha	T-Alpha	Beta	T-Beta	R^2
Mkt	6.93	-0.22	(-0.90)	0.81	(20.17)	0.30
SMB	2.44	0.89	(0.72)	0.16	(5.25)	0.03
HML	5.04	0.66	(0.02)	0.47	(17.87)	0.30
Mom	8.03	5.08	(2.86)	0.29	(6.82)	0.05

mechanically inherit this ability. We show that this is not the case using two approaches. First, we compare the LMP to benchmark models. We find that the LMP alone prices nearly as well as the three Fama-French factors and the

Table X: pricing benchmark factors with the LMP

Table XI
Mean-Standard Deviation Analysis

This table presents the monthly mean, standard deviation, and Sharpe ratios of the Fama-French three factors, momentum factor, LMP, and maximum possible Sharpe ratio from any combination of the Fama-French three factors and momentum factor. We provide the annual Sharpe ratio by multiplying by $\sqrt{12}$. Data are monthly from 1936Q1 to 2009Q4.

	$E[R^f]$	$\sigma[R^f]$	Sharpe Ratio	Annualized Sharpe
Market	0.57	4.30	0.13	0.46
SMB	0.15	2.86	0.05	0.18
HML	0.40	2.75	0.15	0.50
Mom	1.32	6.48	0.20	0.70
LMP	1.92	3.23	0.29	0.99
Max Sharpe			0.35	1.20

regressions. We form portfolios based on ex-ante leverage betas and show that the resulting portfolios have a large spread in average returns. Specifically, at each quarter (the formation date), we sort the universe of Amex, NASDAQ, and NYSE stocks from CRSP based on their estimated leverage betas over the past 10 years (40 quarters), which we call prereanking betas. We compute leverage

Table XI: mean-standard deviation analysis

- The low-minus-high beta spread is about 6.99% per year, with a Sharpe ratio of 0.36.
- The leverage factor prices the beta-sorted portfolio cross-section with high adjusted R^2 .

Economic message: These results strengthen the interpretation that the leverage factor captures funding constraints, not merely a statistical cross-sectional pattern among the original test assets.

Table XII
Leverage Beta Sorts

This table presents sorts of the entire universe of CRSP stocks into portfolios based on preranking leverage betas, found by regressing quarterly firm-level excess returns (over the three-month risk-free rate) on our leverage factor. We use 10 years (40 quarters) for presorting regressions. We exclude the smallest 10% of firms based on market capitalization on formation date and exclude firms that do not have at least 20 quarters of past return data on formation date. We report the mean, standard deviation, Sharpe ratio, postformation leverage beta, and t -statistic as well as the CAPM α and associated t -statistic. We report numbers in percent per year. Leverage betas are multiplied by 100 for convenience. Our holding period is 1973 to 2009.

Panel A: Three Sorted Leverage Portfolios				
	Low	Med	High	High-Minus-Low
$E[R]$	5.17	5.94	7.99	2.82
$\sigma[R]$	19.63	16.99	21.01	13.44
$E[R]/\sigma[R]$	0.26	0.35	0.38	0.21
Leverage β	3.81	7.76	11.39	7.58
$t-\beta$	0.61	1.54	1.86	2.01
CAPM α	-0.53	1.08	2.12	2.64
$t-\alpha$	-0.56	1.38	1.48	1.17

Panel B: Five Sorted Leverage Portfolios

	1 (Low)	2	3	4	5 (High)	5-1
$E[R]$	4.91	5.62	5.81	7.12	8.78	3.87
$\sigma[R]$	22.76	17.80	17.14	18.73	24.06	18.56
$E[R]/\sigma[R]$	0.22	0.32	0.34	0.38	0.36	0.21

Table XII: leverage beta sorts

Table XIII
“Betting against Beta” Portfolios

This table presents sorts of CRSP stocks on 10-year preranking market betas. We lever up low-beta stocks so that each portfolio has a market beta of one. The resulting portfolios comove with funding conditions since leverage-constrained investors will bid up high-beta securities (Frazzini and Pedersen (2013)). See the text for a complete description. FF represents the Fama-French factors, and Mom adds the momentum factor. The holding period is 1968 to 2009. All data are quarterly but returns and Sharpe ratios are presented as annualized numbers (multiplied by 4 and 2, respectively).

Panel A: Time-Series Regressions: $R_{i,t}^e = c_i + \beta_{Lev,i} LevFac_t + \epsilon_{i,t}$					
	$E[R^e]$	Sharpe	$\beta_{Lev} (\times 10^{-2})$	t -stat	R^2
BAB1	10.98	0.46	19.45	2.93	4.90%
BAB2	8.94	0.40	21.71	3.50	6.88%
BAB3	7.29	0.36	16.41	2.91	4.84%
BAB4	6.87	0.35	11.33	2.01	2.38%
BAB5	6.68	0.34	11.67	2.11	2.60%
BAB6	4.67	0.25	12.91	2.41	3.38%
BAB7	5.68	0.30	10.19	1.89	2.10%
BAB8	4.68	0.25	8.90	1.67	1.66%
BAB9	4.29	0.22	3.97	0.72	0.31%
BAB10	3.99	0.20	3.51	0.62	0.23%
1 - 10	6.99	0.36	15.94	2.90	4.82%

Panel B: Cross-Sectional Test Diagnostics

MAPE:	$E[R^e]$	FF	FF, Mom	LevFac	LevFac, No Int
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Table XIII: betting-against-beta portfolios

6 Short summary

- Adrian, Etula, and Muir propose an intermediary-based SDF using broker-dealer leverage shocks.
- The leverage factor is seasonally adjusted quarterly log changes in broker-dealer leverage.
- The economic interpretation is that deleveraging indicates tight funding conditions and high intermediary marginal value of wealth.
- The factor is procyclical and comoves with asset growth, volatility, credit spreads, and financial-sector returns in the expected directions.
- A single leverage factor prices 25 size/book-to-market portfolios, 10 momentum portfolios, and six Treasury bond portfolios with adjusted R^2 around 77% and total MAPE around 1.3%.
- The factor’s price of risk is positive and statistically significant.
- Benchmark models need several factors to achieve similar performance.
- Simulation exercises show that random factors are unlikely to replicate the factor’s high R^2 and low intercept.
- A tradable LMP constructed from benchmark returns performs well in monthly tests and has strong mean-variance properties.
- Leverage-beta sorts and betting-against-beta portfolios support the funding-constraint interpretation.
- The paper’s main contribution is to put financial intermediaries, rather than representative households, at the center of empirical asset pricing.

7 Conclusion

7.1 Core conclusion

The paper shows that broker-dealer leverage shocks contain economically meaningful information about the cross-section of expected returns. A one-factor intermediary leverage model prices a broad collection of

equity and bond portfolios almost as well as standard multifactor benchmarks. This supports the view that intermediary balance-sheet conditions are central to the stochastic discount factor.

7.2 Economic intuition

When funding conditions are loose, broker-dealers expand balance sheets and leverage rises. When funding conditions tighten, broker-dealers delever, asset prices fall, and intermediary marginal value of wealth rises. Assets that covary positively with leverage shocks tend to perform well in good intermediary states and poorly in bad intermediary states, so investors require compensation for holding them.

7.3 Takeaway

The enduring message is that the marginal investor in modern financial markets may not be the representative household. Financial intermediaries trade actively across markets, and their balance-sheet constraints can shape asset prices. Broker-dealer leverage is therefore a useful empirical window into the intermediary SDF.